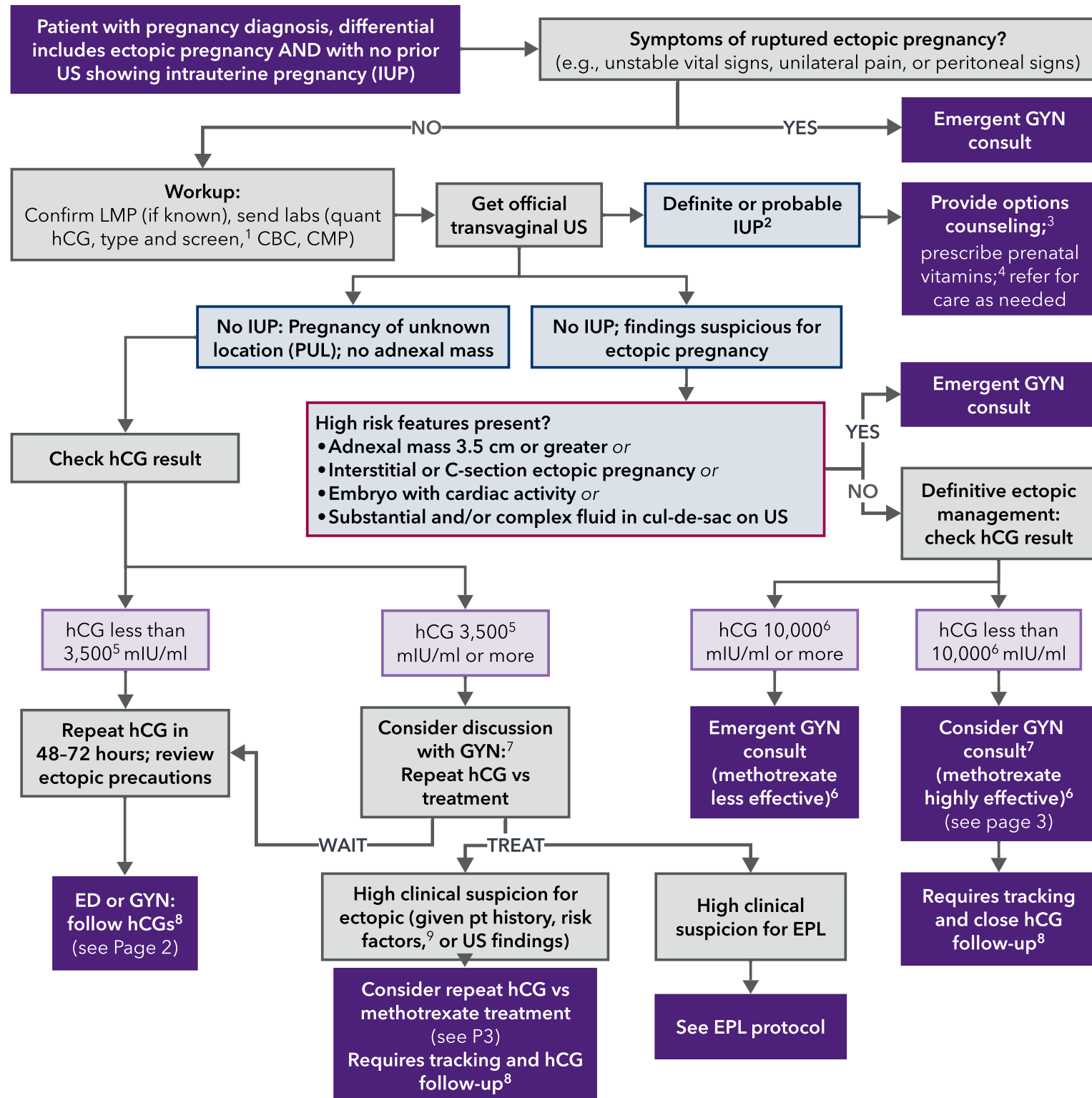


PUL & Ectopic Pregnancy: Work-Up and Management in the Emergency Department *(in states limiting abortion care/referrals)*

Review with collaborating GYN referral/back-up to obtain mutual agreement on protocol prior to use.
PUL is pregnancy of uncertain location.

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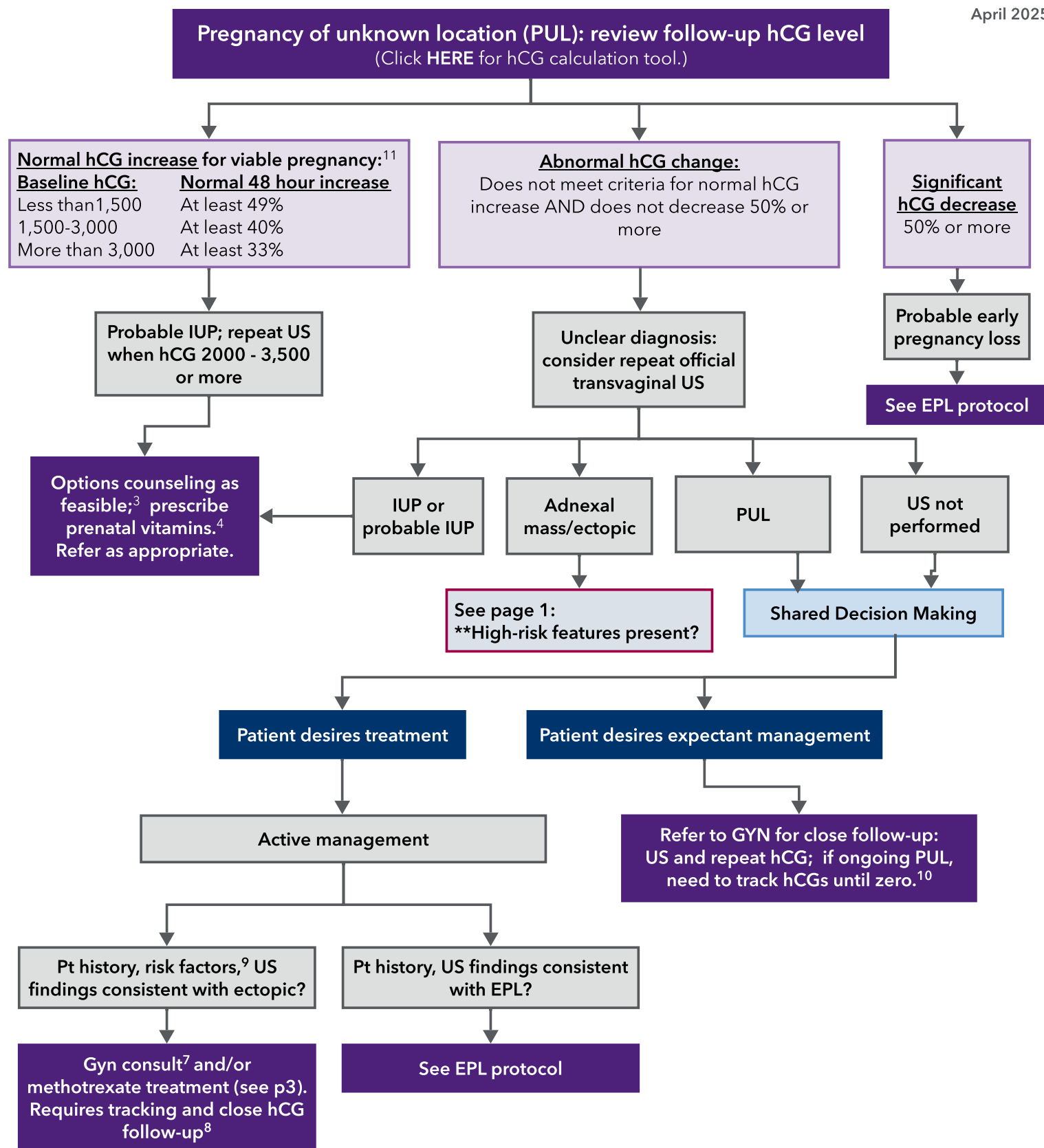


See page 4 for endnotes.

The Reproductive Health Hotline (1 844-REPROHH or 1 844-737-7644)

Free, confidential, nationwide, evidence-based hotline for health care providers. Monday-Friday 6am-5pm PT / 9am-8pm ET.

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- Must have arranged referral relationships for follow-up OR an assigned person in facility to track hCGs until zero.
- Patients must have no absolute contraindications.

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Contraindications to Methotrexate Therapy¹¹

Absolute contraindications

- Ruptured ectopic pregnancy
- Hemodynamic instability
- Intrauterine pregnancy
- Evidence of immunodeficiency
- Moderate to severe anemia, leukopenia, thrombocytopenia
- Active pulmonary disease

Absolute contraindications, continued

- Active peptic ulcer disease
- Clinically important hepatic or renal dysfunction
- Breastfeeding
- Sensitivity to methotrexate
- Inability to participate in follow-up

Relative contraindications (GYN consult)

- Cardiac activity detected on US
- High initial hCG (see below for failure rates by baseline hCG)
- 4.0 cm or larger adnexal mass on TVUS
- Refusal to accept blood transfusion, if needed

Baseline hCG less than 5,000⁶ and no adnexal mass, or mass less than 3.5 cm¹²

Baseline hCG 5,000-9,999⁶ and/or adnexal mass more than 3.5 cm;¹² recommend GYN consult

Day 1 (date of injection)	Perform CBC, AST, ALT, creatinine If within normal limits, MTX IM 50 mg/m ²	Perform CBC, AST, ALT, creatinine If within normal limits, MTX IM 50 mg/m ²
Day 4	Repeat hCG. Day 1 to day 4 rise is expected.	Repeat dose of MTX IM 50 mg/m ² Repeat hCG. Day 1 to day 4 rise is expected.
Day 7	Repeat hCG. Day 4 to Day 7 levels should drop at least 15%. If less than 15% drop: repeat MTX IM 50 mg/m ² (same as above) or refer to surgery. If 15% or more drop, MTX is working. Repeat hCG weekly until zero.	Repeat hCG. If drop between Day 4 and 7 is less than 15%, refer to GYN for repeat MTX IM or surgery. If 15% or more drop, MTX is working. Repeat hCG weekly until zero.

Patient Education

- Avoid folic acid, prenatal vitamins, alcohol, NSAIDs, sexual intercourse and strenuous activity until completion of treatment.
- Mild-moderate abdominal pain, nausea and vomiting are normal.
- Go to ED if severe abdominal pain or dizziness (signs of rupture).
- Must follow hCG weekly until zero.
- Avoid pregnancy for three months (use reliable contraception).
- Stay within 30 minutes of an ED until treatment complete.

Success Rates of Single-Dose Methotrexate IM for Ectopic Pregnancy⁶

Initial hCG level	Success rate
Less than 1,000	98%
1,000 to 1,999	94%
2,000 to 4,999	96%
5,000 to 9,999	85%
10,000 or more	81%

Ectopic Risk Factors

- Previous ectopic pregnancy (10% risk with history of one ectopic, 25% risk with history of two ectopics)
- Pregnancy while intrauterine device in place
- Pelvic inflammatory disease or history of chlamydia or gonorrhea
- Previous fallopian tube surgery or documented fallopian tube pathology
- Infertility and history of fertility treatments
- Age over 35 years
- Cigarette smoking

Interpreting hCG Levels¹¹

hCGs changes over time are variable

- Normal rise does not guarantee normal pregnancy.
- Abnormal levels do not always indicate abnormal pregnancy.
- Consider GYN consultation in strongly desired pregnancies.
- hCGs typically plateau at 10 weeks gestation; hCG trending is not predictive in later pregnancy
- **Normal rate of hCG increase** after 48h for viable pregnancy:
 - Baseline hCG <1,500: at least 49%
 - Baseline hCG 1,500-3,000: at least 40%
 - Baseline hCG >3,000: at least 33%

Early Pregnancy Loss: normal decrease (if less, suggests ectopic)

- After 48 hours: expect at least 21%
- After 4 days: 85% rules out ectopic
- After 7 days: expect at least 60%; 95% rules out ectopic
- In the setting of an IUP of uncertain viability, hCG rise less than 11% is diagnostic of nonviable pregnancy.¹³

See page 4 for endnotes.

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For additional Access Bridge resources, visit bridgetotreatment.org/access-bridge/.

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ENDNOTES:

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- 1 Type and screen for potential cross match. Rh testing and RhoGAM administration are not recommended before 12 weeks' gestation for EPL, medical abortion or procedural abortion ([ACOG 2024](#), [SFP 2022](#)). RhoGAM is recommended for Rh negative patients with ectopic pregnancy. Follow institutional guidance.
- 2 Probable IUP means intrauterine fluid (sometimes labeled as gestational sac) without a yolk sac; a retrospective study found the incidence of ectopic pregnancy in patients with intrauterine fluid collections was very low; follow-up ultrasound in one week can be offered. [Flynn 2023](#).
- 3 [All-Options Talkline](#) 1-888-493-0092; See [options counseling guide](#).
- 4 Folic acid prevents birth defects if taken early in pregnancy ([see CDC](#)).
- 5 The discriminatory zone is the hCG level at which a gestational sac (GS) should be visible on transvaginal US. hCG level of 3,500 is the threshold used by ACOG. ([Connolly 2013](#)).
- 6 If methotrexate used, failure rate based on pre-treatment hCG: 3.7% if <5,000; 15% if 5,000 - 10,000; 19% if $\geq 10,000$ ([Menon 2007](#)).
- 7 If available and clinically appropriate, diagnostic aspiration to rule out EPL may be done prior to methotrexate treatment. [Flynn 2023](#).
- 8 Requires responsible party to track patient and call if no show for follow-labs (by ED case management or confirmed "warm hand-off" to outpatient clinic, GYN, or other referral partner).
- 9 Tailor treatment and follow-up to patient's risk factors for ectopic pregnancy: 50% if IUD in place at time of pregnancy, 10% if history of prior ectopic, 10-15% with prior tubal surgery. Other risk factors: estimated GA by LMP over 42 days if PUL on ultrasound, history of PID, chlamydia or gonorrhea. [ACOG 2018](#).
- 10 In setting of PUL and suspected ectopic pregnancy, need to follow hCGs until zero, [ACOG 2018](#). Use shared decision-making with patient weighing relative risks, costs, and availability of follow-up. 7% of PULs end up being ectopic pregnancies. [Park 2023](#). From
- 11 Tubal Ectopic Pregnancy, [ACOG 2018](#).
- 12 Ectopic pregnancies are more likely to rupture over 3.5cm; recommend GYN consult prior to treatment. [Barash 2023](#).
- 13 Increase of less than 11% is 100% diagnostic of nonviable pregnancy ([Society of Family Planning Medication Management of Early Pregnancy Loss](#))

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